# Connected Bodies (Pulleys)

Exam code: 9MA0

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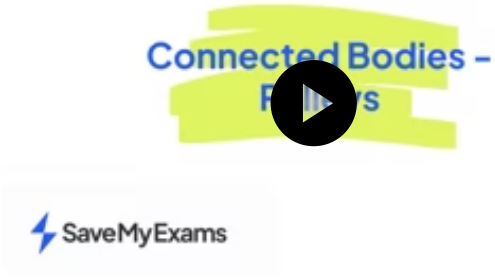


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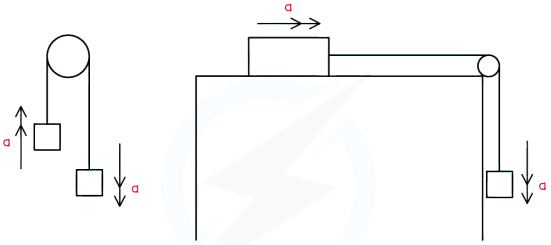
## Connected bodies (pulleys)

### What is a pulley?

- A **pulley** is a wheel like device that rotates as a string passes over it allowing motion of any particles attached to the string
- Pulleys allow a (inextensible) **string** to change its **orientation**
- In A level mathematical models, pulleys will always be **smooth** and **light**, so there is no friction involved at the pulley and its mass is negligible
- A **peg** is similar to a pulley but is a fixed point that a particle can be suspended from (like a nail in a wall)



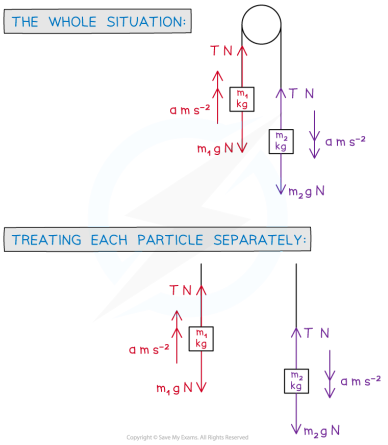
EXAMPLES OF PARTICLES CONNECTED BY A LIGHT INEXTENSIBLE STRING PASSING OVER A SMOOTH PULLEY.
NOTE HOW, IN BOTH CASES, THE PARTICLES ARE TRAVELLING IN DIFFERENT DIRECTIONS.

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

### How do I solve pulley questions?

- If a particle is in **motion** in the direction being considered then **Newton's Laws of Motion** apply so use "*F = ma*" (N2L)
- For **constant acceleration** the '*suvat*' equations could be involved
- Step 1. Draw a series of diagrams
  - Label the forces and the positive direction of motion for each particle.
  - Colour coding forces acting on each particle may help




- Step 2. Write equations of motion, using "*F = ma*"
  - **Equations 1** and **2**: Treating each particle separately

$$(\uparrow)\ T - m_1 g = m_1 a$$

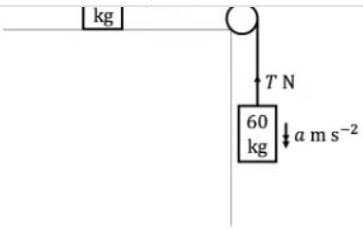
$$(\downarrow)\ m_2 g - T = m_2 a$$

- Step 3. Solve the relevant equation(s) and answer the question
  - Some trickier problems may lead to simultaneous equations

## Worked Example

A block of mass 25 kg lies on a smooth horizontal table. A light, inextensible string passes over a smooth pulley attached to the edge of the table. The string connects the 25 kg block to another, of mass 60 kg, that hangs freely as shown in the diagram below.

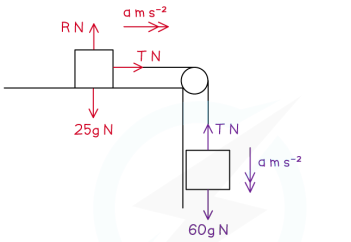
The system is released from rest.



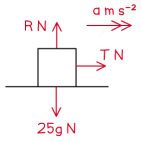

Find the tension, $T$ N , in the string and the acceleration, a m $s^{-2}$ ,of the system.

**Answer:**

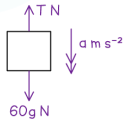
Step 1: Sketch appropriate diagrams (You can add to the one given)




25 kg block only:




60 kg block only:

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Step 2: Write equations for each

(→) $T = 25a$

(↓) $60g - T = 60a$

(We are not concerned with R so do not need to consider the vertical direction)

Step 3: Solve the appropriate equation(s)

Both equations involve T and a so solve simultaneously

$T = 25a$ *

$60g - T = 60a$ **

Substitute * into ** $60g - 25a = 60a$

$85a = 60g$

$a = \frac{12}{17}g = 6.91764...$ $g = 9.8$

$T = 25 \times 6.91764 = 172.941...$

∴ $T = 173$ N (3 s.f.)

$a = 6.92$ m $s^{-2}$ (3 s.f.)

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## Examiner Tips and Tricks

- Sketch a diagram or add to a diagram given in a question.
- All pulleys are smooth and light; in many questions the pulley itself can be ignored.
- In pulley questions the particles will be moving in different directions - so each particle needs to be considered separately.

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